

A Rare-Jump Obstruction for Wasserstein-Penalty Dynamic Risk Scaling

June 24, 2026

Abstract

Backhoff-Veraguas and Zhang asked in Remark 2.4 of their paper on specific Wasserstein divergence for the scaling limit of the dynamic risk/robust hedging problem whose one-step penalty is $\alpha_1(q) = \pm W_1(q, P)^p$. This note gives a corrected and deliberately modest contribution to that question. It does not identify the full scaling limit and therefore is not a solution of the open problem. Instead, for the Brownian reference model, it proves a no-go result for a natural unrestricted positive-sign formulation: every normalization $a_N = o(N^{p-1})$, in particular the fixed-diffusion normalization $a_N = N^{p/2-1}$, admits martingale laws with rare jumps of diverging size and vanishing normalized W_1^p chain cost. Consequently bounded continuous escape payoffs have normalized robust values converging to their maximal value. For $p \geq 2$ this contradicts the naive continuous-path variational candidate obtained by inserting the specific Wasserstein divergence as a penalty. The negative-sign penalty is also shown to be infinite already at the zero payoff on the usual unbounded state space. Finally, a proposed repair based only on an $r > 1$ diffusive moment bound is corrected: such a bound does not exclude the rare-jump construction when $1 < r < 2$.

1 Status and scope

The result below is *partial progress*, not a complete solution of the scaling problem in Remark 2.4 of [1]. The paper [1] proves a scaling limit for the chain-constructed Wasserstein discrepancy between fixed continuous martingale laws. The open question concerns instead the scaling limit of the associated multiperiod dynamic risk/robust hedging optimization. We prove that, in the unrestricted Brownian-reference model, the fixed-law scaling cannot simply be promoted to the corresponding robust optimization over all discrete-time martingale laws.

The obstruction is elementary. A jump of size M_N inserted with probability of order $1/N$ at each time step changes terminal escape payoffs with order-one probability. But the one-step W_1 displacement is only of order M_N/N , and after taking the p th power and summing over N steps its cost is of order $M_N^p N^{1-p}$. Hence every normalization $a_N = o(N^{p-1})$ still allows $M_N \rightarrow \infty$ with vanishing normalized cost.

2 Discrete-time setup

Let P_N be the law of the Gaussian martingale random walk

$$X_0 = 0, \quad X_i - X_{i-1} \sim N(0, 1/N) \quad \text{independently,} \quad i = 1, \dots, N.$$

For a law $Q \in \mathcal{P}_1(\mathbb{R}^N)$ write $Q_i^{x_{1:i-1}}$ for a regular conditional law of X_i given $X_{1:i-1} = x_{1:i-1}$, and write

$$P_i^{x_{1:i-1}} = N(x_{i-1}, 1/N).$$

We use the additive chain discrepancy

$$D_{N,p}(Q\|P_N) := \mathbb{E}^Q \left[\sum_{i=1}^N W_1(Q_i^{X_{1:i-1}}, P_i^{X_{1:i-1}})^p \right], \quad (1)$$

with the convention $D_{N,p} = +\infty$ if the displayed quantity is not well-defined. Let $\mathcal{M}_N$ denote the set of laws on $\mathbb{R}^N$ under which $(X_i)_{i=0}^N$ is an integrable martingale starting from 0. Let $\hat{X}^N$ be the continuous linear interpolation of $(X_i)_{i=0}^N$.

For a bounded Borel functional F on $C([0, 1]; \mathbb{R})$ and a normalization $a_N > 0$, define the martingale-dual value

$$V_N^a(F) := \sup_{Q \in \mathcal{M}_N} \left\{ \mathbb{E}^Q[F(\hat{X}^N)] - a_N D_{N,p}(Q\|P_N) \right\}. \quad (2)$$

This is the dual expression that appears formally after robust hedging by predictable stochastic integrals.

We shall also use the positive-sign dynamic risk functional

$$\rho_N(f) := \sup_{Q \in \mathcal{P}_1(\mathbb{R}^N)} \left\{ \mathbb{E}^Q[f] - D_{N,p}(Q\|P_N) \right\}. \quad (3)$$

For $p \geq 1$ this is the usual convex dual form; for $0 < p < 1$ the penalty is not convex, but the formula still defines the robust functional used below. If $\mathcal{H}_N$ is any class of predictable strategies containing 0 and such that $(H \cdot X)_N$ is integrable with zero expectation under every martingale law used in the proof, set

$$\Pi_N^a(F) := a_N \inf_{H \in \mathcal{H}_N} \rho_N \left(a_N^{-1} F(\hat{X}^N) - (H \cdot X)_N \right). \quad (4)$$

For example, the class of bounded predictable strategies satisfies this condition for all laws in $\mathcal{M}_N$ with finite first moments.

Lemma 2.1 (The dual lower bound needs no minimax). *For every bounded F and every N ,*

$$\Pi_N^a(F) \geq V_N^a(F).$$

If additionally $0 \leq F \leq 1$, then $\Pi_N^a(F) \leq 1$ and $V_N^a(F) \leq 1$.

Proof. Fix $Q \in \mathcal{M}_N$ and $H \in \mathcal{H}_N$. By (3),

$$a_N \rho_N \left(a_N^{-1} F(\hat{X}^N) - (H \cdot X)_N \right) \geq \mathbb{E}^Q[F(\hat{X}^N)] - a_N \mathbb{E}^Q[(H \cdot X)_N] - a_N D_{N,p}(Q\|P_N).$$

The stochastic integral has mean zero under Q , so the right-hand side is $\mathbb{E}^Q[F(\hat{X}^N)] - a_N D_{N,p}(Q\|P_N)$. Taking the infimum over H and then the supremum over $Q \in \mathcal{M}_N$ gives $\Pi_N^a(F) \geq V_N^a(F)$.

If $0 \leq F \leq 1$, then $D_{N,p} \geq 0$, so $V_N^a(F) \leq 1$. Taking $H = 0$ in (4) gives

$$\Pi_N^a(F) \leq a_N \rho_N(a_N^{-1} F(\hat{X}^N)) = \sup_{Q \in \mathcal{P}_1(\mathbb{R}^N)} \left\{ \mathbb{E}^Q[F(\hat{X}^N)] - a_N D_{N,p}(Q\|P_N) \right\} \leq 1.$$

□

3 The fixed-diffusion normalization

The following standard calculation explains why the specific-Wasserstein scaling suggests $a_N = N^{p/2-1}$.

Lemma 3.1 (Fixed Gaussian volatility). *Let Q_N^σ be the law of the Gaussian martingale random walk with increments $N(0, \sigma^2/N)$, where $\sigma \geq 0$ is constant. Then*

$$D_{N,p}(Q_N^\sigma \| P_N) = N^{1-p/2} \left(\frac{2}{\pi} \right)^{p/2} |\sigma - 1|^p. \quad (5)$$

Consequently a normalization producing a finite non-zero cost for each fixed $\sigma \neq 1$ must satisfy

$$a_N N^{1-p/2} \rightarrow a \in (0, \infty),$$

that is, a_N must be asymptotic to a positive multiple of $N^{p/2-1}$.

Proof. At each time step the two conditional laws are $N(x, \sigma^2/N)$ and $N(x, 1/N)$. By translation invariance and the monotone coupling of one-dimensional centered Gaussians,

$$W_1(N(x, \sigma^2/N), N(x, 1/N)) = \mathbb{E} \left| \frac{\sigma Z}{\sqrt{N}} - \frac{Z}{\sqrt{N}} \right| = N^{-1/2} |\sigma - 1| \left(\frac{2}{\pi} \right)^{1/2},$$

where $Z \sim N(0, 1)$. Summing the p th power of this quantity over N steps gives (5). $\square$

4 Rare jumps at vanishing normalized cost

Fix $\lambda > 0$ and a deterministic sequence $M_N \rightarrow \infty$. For $N > \lambda$, define a Markov law $Q_N^{\lambda, M} \in \mathcal{M}_N$ by the transition rule

$$X_i = \begin{cases} X_{i-1} + Z_i/\sqrt{N}, & \text{with probability } 1 - \lambda/N, \\ X_{i-1} + M_N, & \text{with probability } \lambda/(2N), \\ X_{i-1} - M_N, & \text{with probability } \lambda/(2N), \end{cases} \quad (6)$$

where the Z_i are independent standard Gaussians, independent of the jump indicators. The conditional mean of X_i given the past is X_{i-1} , so $Q_N^{\lambda, M}$ is a martingale law.

Lemma 4.1 (Rare-jump cost). *For every $p > 0$ there is a constant $C_{p,\lambda} < \infty$ such that, for all sufficiently large N ,*

$$D_{N,p}(Q_N^{\lambda, M} \| P_N) \leq C_{p,\lambda} M_N^p N^{1-p}. \quad (7)$$

Consequently, if $a_N N^{1-p} M_N^p \rightarrow 0$, then

$$a_N D_{N,p}(Q_N^{\lambda, M} \| P_N) \rightarrow 0.$$

Proof. At a fixed state x , the transition in (6) is the mixture

$$\left(1 - \frac{\lambda}{N}\right) N(x, 1/N) + \frac{\lambda}{2N} \delta_{x+M_N} + \frac{\lambda}{2N} \delta_{x-M_N}.$$

Couple the no-jump part identically with the reference law $N(x, 1/N)$. On the jump event, couple the atom $x \pm M_N$ to an independent reference Gaussian $x + Z/\sqrt{N}$. This gives

$$W_1((Q_N^{\lambda, M})_i^{x_{1:i-1}}, P_i^{x_{1:i-1}}) \leq \frac{\lambda}{N} \left(M_N + \frac{\mathbb{E}|Z|}{\sqrt{N}} \right).$$

Since $M_N \rightarrow \infty$, the right-hand side is at most $C_\lambda M_N/N$ for all large N . Taking the p th power and summing over N steps proves (7). $\square$

The next lemma records the effect of the same law on terminal escape payoffs.

Lemma 4.2 (Escape probability). *Let $0 \leq \varphi \leq 1$ be continuous and satisfy*

$$\lim_{|x| \rightarrow \infty} \varphi(x) = 1.$$

For $F(\omega) = \varphi(\omega_1)$ and every fixed $\lambda > 0$,

$$\liminf_{N \rightarrow \infty} \mathbb{E}^{Q_N^{\lambda, M}} [F(\hat{X}^N)] \geq \mathbb{P}(S_\lambda \neq 0),$$

where $S_\lambda = Y_+ - Y_-$ and Y_+, Y_- are independent Poisson random variables of mean $\lambda/2$. Moreover, $\mathbb{P}(S_\lambda \neq 0) \rightarrow 1$ as $\lambda \rightarrow \infty$.

Proof. Let $Y_{N,+}$ and $Y_{N,-}$ be the numbers of upward and downward jumps. The vector $(Y_{N,+}, Y_{N,-})$ is multinomial with cell probabilities $\lambda/(2N)$, $\lambda/(2N)$, and $1 - \lambda/N$, hence

$$(Y_{N,+}, Y_{N,-}) \Rightarrow (Y_+, Y_-),$$

with Y_+, Y_- independent Poisson($\lambda/2$). Thus

$$S_N := Y_{N,+} - Y_{N,-} \Rightarrow S_\lambda.$$

Write

$$X_N = M_N S_N + G_N,$$

where G_N is the sum of the Gaussian increments occurring on no-jump steps. The sequence (G_N) is tight, since $\mathbb{E}[G_N^2] \leq 1$. On the event $\{|S_N| \geq 1\}$, the term $M_N S_N$ diverges in absolute value in probability, while G_N remains tight. Therefore

$$\liminf_{N \rightarrow \infty} \mathbb{E}^{Q_N^{\lambda, M}} [\varphi(X_N)] \geq \lim_{N \rightarrow \infty} \mathbb{P}(|S_N| \geq 1) = \mathbb{P}(S_\lambda \neq 0).$$

Finally S_λ has the Skellam distribution and

$$\mathbb{P}(S_\lambda = 0) = e^{-\lambda} I_0(\lambda),$$

where I_0 is the modified Bessel function. This tends to 0 as $\lambda \rightarrow \infty$. $\square$

Theorem 4.3 (Zero-cost escape for the positive-sign problem). *Let $p > 0$ and let $a_N > 0$ satisfy*

$$a_N = o(N^{p-1}). \tag{8}$$

Let $0 \leq \varphi \leq 1$ be continuous with $\varphi(x) \rightarrow 1$ as $|x| \rightarrow \infty$, and put $F(\omega) = \varphi(\omega_1)$. Then

$$\lim_{N \rightarrow \infty} V_N^a(F) = 1.$$

For every admissible class of strategies satisfying the condition stated before (4), one also has

$$\lim_{N \rightarrow \infty} \Pi_N^a(F) = 1.$$

In particular this applies to the fixed-diffusion normalization $a_N = N^{p/2-1}$ for every $p > 0$.

Proof. Condition (8) is equivalent to $b_N := a_N N^{1-p} \rightarrow 0$. Choose a deterministic $M_N \rightarrow \infty$ such that $b_N M_N^p \rightarrow 0$. Lemma 4.1 gives

$$a_N D_{N,p}(Q_N^{\lambda, M} \| P_N) \rightarrow 0$$

for every fixed $\lambda > 0$. Lemma 4.2 therefore yields

$$\liminf_{N \rightarrow \infty} V_N^a(F) \geq \mathbb{P}(S_\lambda \neq 0).$$

Letting $\lambda \rightarrow \infty$ gives $\liminf_N V_N^a(F) \geq 1$. Since $0 \leq F \leq 1$, Lemma 2.1 gives $V_N^a(F) \leq 1$ and also sandwiches $\Pi_N^a(F)$ between $V_N^a(F)$ and 1. Hence both limits equal 1.

Finally, $N^{p/2-1} = o(N^{p-1})$ because $N^{p/2-1}/N^{p-1} = N^{-p/2} \rightarrow 0$. $\square$

Corollary 4.4 (Incompatibility of the two requirements). *The normalization $a_N \asymp N^{p/2-1}$ is the only normalization that gives a finite non-zero penalty for fixed Gaussian volatility mismatches, but under that same normalization the unrestricted positive-sign robust problem has zero-cost escape for bounded terminal payoffs. More generally, every normalization $a_N = o(N^{p-1})$ has the zero-cost escape property of Theorem 4.3. A normalization large enough to fall outside this regime necessarily destroys the fixed-diffusion scale: if $a_N N^{1-p}$ does not tend to 0, then along a subsequence the cost in (5) diverges for every fixed $\sigma \neq 1$.*

Proof. The first statement is Lemma 3.1 together with Theorem 4.3. For the last statement, write

$$a_N D_{N,p}(Q_N^\sigma \| P_N) = (a_N N^{1-p}) N^{p/2} \left(\frac{2}{\pi}\right)^{p/2} |\sigma - 1|^p.$$

If $a_N N^{1-p}$ is bounded below by a positive constant along some subsequence, this expression diverges along that subsequence. $\square$

5 Comparison with the naive continuous specific-Wasserstein candidate

The fixed-law specific-Wasserstein formula suggests, in the Brownian reference case, the continuous-path candidate

$$\mathcal{V}_p(F) := \sup_Q \left\{ \mathbb{E}^Q[F(X)] - c_p \mathbb{E}^Q \int_0^1 |\sigma_t - 1|^p dt \right\}, \quad c_p := \left(\frac{2}{\pi}\right)^{p/2}, \quad (9)$$

where the supremum is over continuous martingale laws admitting a representation

$$dX_t = \sigma_t dB_t, \quad X_0 = 0,$$

with progressively measurable $\sigma_t \geq 0$. The following proposition shows that, for $p \geq 2$, (9) cannot be the scaling limit of the unrestricted discrete robust values at the fixed-diffusion normalization.

Proposition 5.1 (Gap for $p \geq 2$). *Assume $p \geq 2$. For $R > 0$, let $\varphi_R : \mathbb{R} \rightarrow [0, 1]$ be continuous with*

$$\varphi_R(x) = 0 \quad \text{for } |x| \leq R, \quad \varphi_R(x) = 1 \quad \text{for } |x| \geq 2R,$$

and set $F_R(\omega) = \varphi_R(\omega_1)$. With $a_N = N^{p/2-1}$,

$$\lim_{N \rightarrow \infty} V_N^a(F_R) = \lim_{N \rightarrow \infty} \Pi_N^a(F_R) = 1,$$

but

$$\mathcal{V}_p(F_R) \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Therefore the unrestricted discrete problem does not converge to the naive continuous variational problem (9) for these payoffs.

Proof. The discrete assertion is Theorem 4.3. We prove the continuous estimate. For a candidate continuous martingale law Q , define

$$A(Q) := \mathbb{E}^Q \int_0^1 |\sigma_t - 1|^p dt.$$

By Ito isometry and $u^2 \leq 2 + 2|u - 1|^2$ for $u \geq 0$,

$$\mathbb{E}^Q[X_1^2] = \mathbb{E}^Q \int_0^1 \sigma_t^2 dt \leq 2 + 2\mathbb{E}^Q \int_0^1 |\sigma_t - 1|^2 dt.$$

Since $p \geq 2$, Jensen's inequality on the probability space $([0, 1] \times \Omega, dt \otimes Q)$ gives

$$\mathbb{E}^Q \int_0^1 |\sigma_t - 1|^2 dt \leq A(Q)^{2/p}.$$

Thus, using $\varphi_R \leq \mathbf{1}_{\{|x| > R\}}$ and Markov's inequality,

$$\mathbb{E}^Q[F_R(X)] \leq \frac{2 + 2A(Q)^{2/p}}{R^2}.$$

Consequently

$$\mathbb{E}^Q[F_R(X)] - c_p A(Q) \leq \frac{2}{R^2} + \frac{2A(Q)^{2/p}}{R^2} - c_p A(Q).$$

If $p = 2$, the right-hand side is at most $2/R^2$ for all sufficiently large R . If $p > 2$, then $q := 2/p < 1$, and

$$\sup_{A \geq 0} \left\{ \frac{2A^q}{R^2} - c_p A \right\} \rightarrow 0 \quad \text{as } R \rightarrow \infty,$$

because the maximum of a sublinear function with coefficient $2/R^2$ minus a fixed positive linear function tends to 0. Taking the supremum over Q proves $\limsup_{R \rightarrow \infty} \mathcal{V}_p(F_R) \leq 0$. Since the Brownian reference law has zero penalty and $\mathbb{E}[F_R(B)] \geq 0$, the values are nonnegative up to a quantity tending to 0, and the limit is 0. $\square$

Remark 5.2 (Why the comparison is stated for $p \geq 2$). *For $0 < p < 2$, the continuous candidate (9) is itself non-coercive for terminal escape payoffs. Indeed, choose $p < q < 2$ and let*

$$\sigma_t^{(L)} = L \mathbf{1}_{[0, L^{-q}]}(t) + \mathbf{1}_{(L^{-q}, 1]}(t).$$

Then

$$\int_0^1 |\sigma_t^{(L)} - 1|^p dt = L^{-q} |L - 1|^p \rightarrow 0,$$

whereas

$$\text{Var}(X_1^{(L)}) = L^{-q} L^2 + 1 - L^{-q} \rightarrow \infty.$$

Hence $\mathbb{E}[\varphi_R(X_1^{(L)})] \rightarrow 1$ for every fixed R , and $\mathcal{V}_p(F_R) = 1$. Thus the tail payoff F_R separates the discrete unrestricted problem from the naive continuous candidate only in the coercive range $p \geq 2$.

6 The negative-sign penalty

The negative-sign version is ill-posed on the unbounded state space. Already in one step, if the penalty is $\alpha_1(q) = -W_1(q, P)^p$, then the dual risk functional takes the form

$$\rho_1^-(f) = \sup_{q \in \mathcal{P}_1(\mathbb{R})} \left\{ \int f dq + W_1(q, P)^p \right\}.$$

Proposition 6.1 (Infinite at zero payoff). *Let $P \in \mathcal{P}_1(\mathbb{R})$ and $p > 0$. Then*

$$\rho_1^-(0) = +\infty.$$

Consequently the unrestricted dynamic risk measure built from the negative one-step penalty is infinite before any scaling limit is considered.

Proof. Take $q_m = \delta_m$. By the triangle inequality,

$$W_1(\delta_m, P) \geq m - \int |x| P(dx) \rightarrow \infty.$$

Therefore $\rho_1^-(0) \geq W_1(\delta_m, P)^p \rightarrow \infty$. $\square$

7 A correction concerning tail restrictions

The earlier draft suggested that a conditional diffusive moment bound of order $r > 1$ rules out the rare-jump construction. This is false for $1 < r < 2$.

Proposition 7.1 (An $r > 1$ moment bound is not enough). *Fix $1 < r < 2$ and $\lambda > 0$. There are choices $M_N \rightarrow \infty$ with $M_N = o(\sqrt{N})$ such that the rare-jump law (6) satisfies*

$$\mathbb{E}^{Q_N^{\lambda, M}} [|X_i - X_{i-1}|^r \mid X_{1:i-1}] \leq CN^{-r/2} \quad (10)$$

for all i, N , while still having vanishing cost at the fixed-diffusion normalization $a_N = N^{p/2-1}$ and producing the escape effect of Theorem 4.3.

Proof. For the transition (6),

$$\mathbb{E} [|X_i - X_{i-1}|^r \mid X_{1:i-1}] = \left(1 - \frac{\lambda}{N}\right) N^{-r/2} \mathbb{E}|Z|^r + \frac{\lambda}{N} M_N^r.$$

Choose $M_N = N^\beta$ with

$$0 < \beta \leq \frac{1}{r} - \frac{1}{2}.$$

This is possible because $r < 2$. Then $M_N \rightarrow \infty$, $M_N = o(\sqrt{N})$, and

$$\frac{M_N^r}{N} = N^{\beta r - 1} \leq N^{-r/2}.$$

This proves (10). Since $M_N = o(\sqrt{N})$, Lemma 4.1 gives

$$N^{p/2-1} D_{N,p}(Q_N^{\lambda, M} \| P_N) \leq C_{p,\lambda} \left(\frac{M_N}{\sqrt{N}} \right)^p \rightarrow 0,$$

and the escape conclusion follows from Lemma 4.2. $\square$

Remark 7.2 (What kind of restriction would exclude the example). *A uniform conditional second moment bound of diffusive order,*

$$\mathbb{E}[|X_i - X_{i-1}|^2 \mid X_{1:i-1}] \leq C/N,$$

would prevent the above construction from having $M_N \rightarrow \infty$ with jump probability λ/N , because it would require $M_N^2/N = O(1/N)$, i.e. $M_N = O(1)$. More generally, a genuine uniform tail condition, rather than merely an L^r bound with $r < 2$, is needed to recover compactness at the diffusive scale.

8 Conclusion

The corrected conclusion is as follows.

- This note does not solve the full open problem in Remark 2.4 of [1]; it gives partial progress in the form of an obstruction.
- The fixed-diffusion calculation forces the normalization $N^{p/2-1}$ if one wants the specific-Wasserstein penalty for fixed continuous martingale laws.
- In the unrestricted positive-sign robust problem, that normalization is too small to prevent rare jumps. In fact every $a_N = o(N^{p-1})$ gives zero-cost escape for bounded continuous terminal payoffs.

- For $p \geq 2$, this proves that the unrestricted discrete robust values cannot converge to the naive continuous specific-Wasserstein variational problem over continuous martingales.
- For $0 < p < 2$, the same tail payoffs do not separate the discrete values from the naive continuous candidate because the latter is itself non-coercive under volatility spikes.
- The negative-sign penalty is infinite at the zero payoff on the unbounded state space.

Thus any positive-sign scaling theorem for the dynamic risk/robust hedging problem must either impose additional compactness or tail restrictions on the admissible discrete-time laws, or else identify a limiting object that incorporates the escape/relaxation effects exposed here. The present note does not carry out that second task.

References

- [1] J. Backhoff-Veraguas and X. Zhang, *Specific Wasserstein divergence between continuous martingales*, Mathematics of Operations Research, 2026. DOI: [10.1287/moor.2025.0896](https://doi.org/10.1287/moor.2025.0896). See also arXiv:2404.19672.